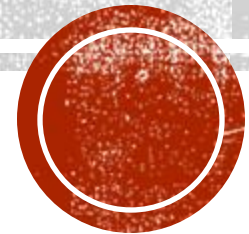


CLOUD COMPUTING

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SYLLABUS

1. Introduction to Cloud Computing
2. Cloud Computing Technologies
3. Networking in Cloud
4. Storage Solutions in the Cloud
5. Cloud Security and Compliance
6. Infrastructure as Code
7. Virtualization and Containerization
8. DevOps and Cloud



WEEK 2: CLOUD COMPUTING TECHNOLOGIES

- **Learning Objectives:**

- Understand what is cloud infrastructure
- Understand what are cloud infrastructure components
- Understand difference between vertical and horizontal scaling
- Introduce Amazon Web Services (AWS)



WHAT IS CLOUD INFRASTRUCTURE?

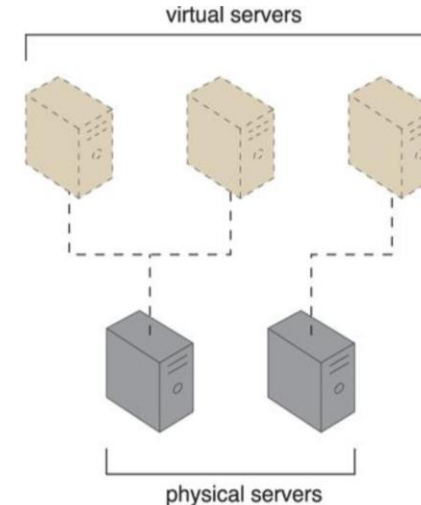
- Cloud infrastructure mechanisms are foundational building blocks of cloud environments that establish primary artifacts to form the basis of fundamental cloud technology architecture.
- Virtual resources mirror those of physical infrastructure, and they include components like memory, network switches, servers, and storage clusters. They are required to create applications that users can access through the cloud or retrieve via the internet, telecom services, and wide-area networks (WANs). The cloud infrastructure approach offers benefits like greater flexibility, scalability, and lower cost of ownership.
- Cloud infrastructure is available for private, public, and hybrid cloud systems. It can also be rented through cloud providers and via several cloud infrastructure delivery models.



COMPONENTS OF CLOUD INFRA (1)

- Compute

- The compute layer of cloud infrastructure is responsible for running and managing virtual machines, containers, and other workloads. It employs virtualization technology to create virtualized resources that can be dynamically allocated and reconfigured.
 - A virtual server/virtual machine: a form of virtualization software that emulates a physical server. Virtual servers are used by cloud providers to share the same physical server with multiple cloud consumers by providing cloud consumers with individual virtual server instances.
 - Containers: lightweight units that package applications and their dependencies, enabling efficient deployment and scaling (e.g., Docker, Kubernetes).
 - Serverless computing: run code without managing the underlying servers, automatically scaling based on demand



COMPONENTS OF CLOUD INFRA (2)

- **Storage:** the cloud storage device mechanism represents storage devices that are designed specifically for cloud-based provisioning. Instances of these devices can be virtualized, similar to how physical servers can spawn virtual server images.
 - **Object Storage:** Stores data as objects, ideal for unstructured data (e.g., images, videos) and scalable storage (e.g., Amazon S3, Google Cloud Storage).
 - **Block Storage:** Provides raw storage volumes for applications requiring low-latency access, often used with virtual machines (e.g., AWS EBS).
 - **File Storage:** Shared file systems accessible over the network, suitable for applications needing file-level access (e.g., Amazon EFS)
- **Networking:** allows users to connect to and deliver hybrid and cloud-native applications with low-latency.
 - **Logical Network Perimeter:** Isolated environments within public clouds that provide secure resource management. E.g. VPC in AWS
 - **Load Balancers:** Distribute network traffic across multiple servers to ensure high availability and performance.
 - **Virtual Firewalls and Security Groups:** Control access to resources by managing incoming and outgoing traffic.



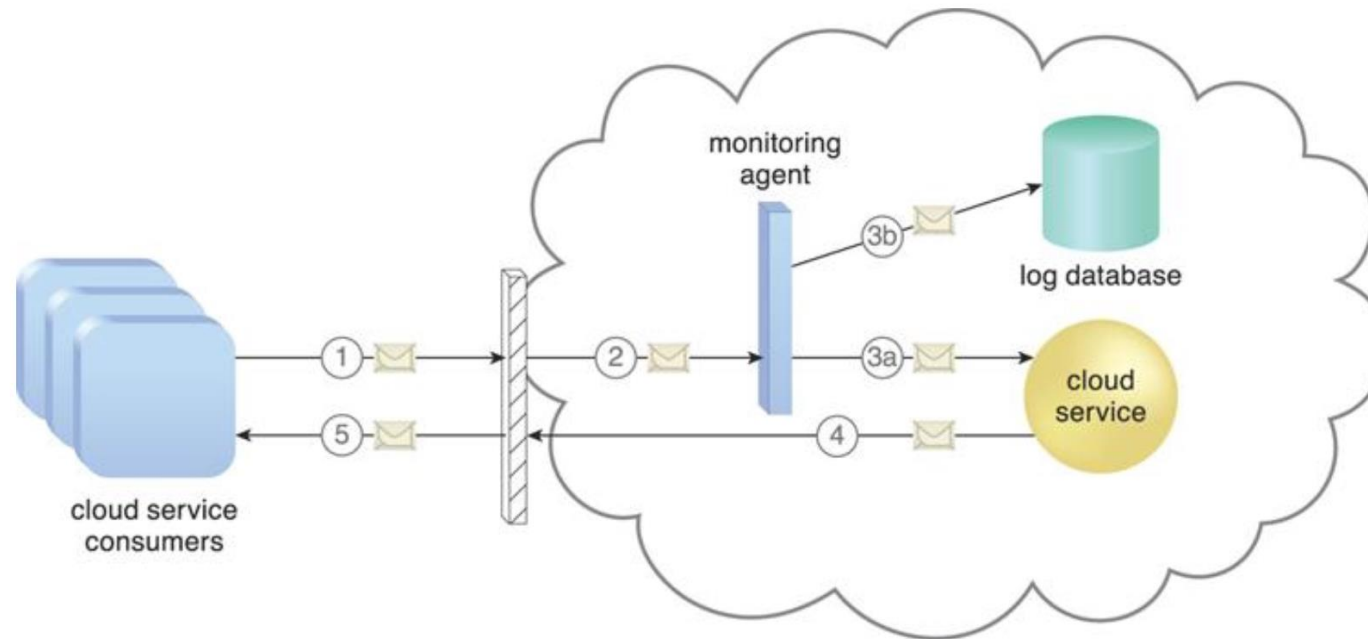
COMPONENTS OF CLOUD INFRA (3)

- Database Storage: cloud storage device mechanisms based on database storage interfaces typically support a query language in addition to basic storage operations.
 - Relational Database: challenges with cloud-based relational databases commonly pertain to scaling and performance. Scaling a relational cloud storage device vertically can be more complex and cost-ineffective than horizontal scaling. Databases with complex relationships and/or containing large volumes of data can be afflicted with higher processing overhead and latency, especially when accessed remotely via cloud services.
 - Non-relational database (NoSQL storage): cloud providers often offer non-relational storage that provides scalability and availability of stored data over multiple server environments. However, many non-relational storage mechanisms are proprietary and therefore can severely limit data portability.



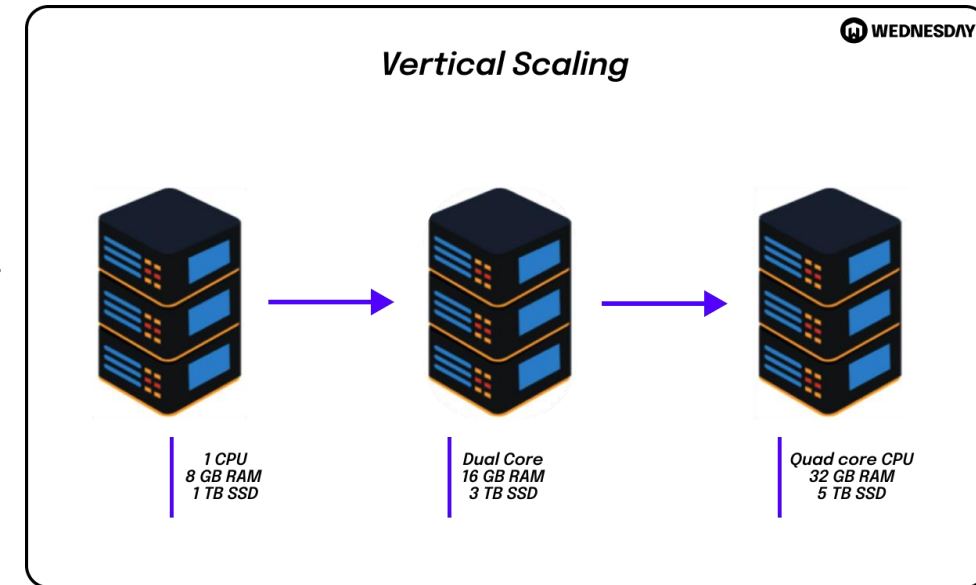
COMPONENTS OF CLOUD INFRA (4)

- Cloud usage monitor mechanism: is a lightweight and autonomous software program responsible for collecting and processing IT resource usage data.
 - Monitoring agent is an intermediary, event-driven program that exists as a service agent and resides along existing communication paths to transparently monitor and analyze dataflows



SCALING IN CLOUD COMPUTING (1)

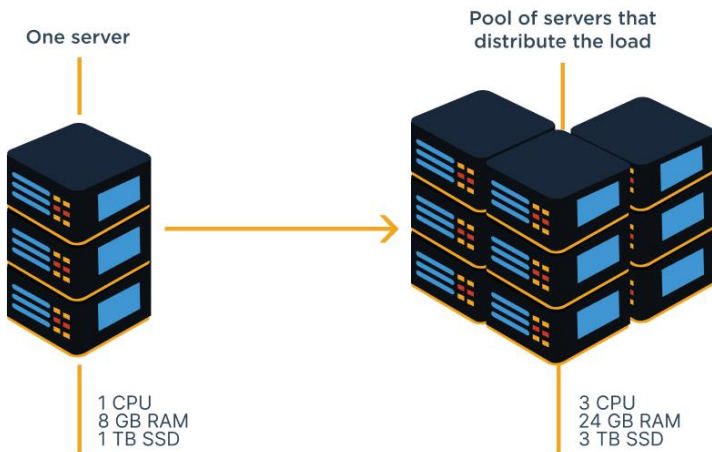
- **Scaling in Cloud Computing** refers to the ability to adjust resources in a cloud environment to meet changing demands. There are two primary types of scaling: **vertical scaling** and **horizontal scaling**. Each approach has its own advantages and use cases.
- **Vertical Scaling (Scaling Up/Down):** Involves adding or removing resources (like CPU, RAM, or storage) to a single server or instance to increase its capacity. For example, upgrading a virtual machine from 4 GB RAM to 8 GB RAM or increasing its CPU cores.
 - **Advantages:** Simplicity as it often requires no changes to the application architecture. Immediate performance improvement without the need for load balancing.
 - **Disadvantages:** Limited by the maximum capacity of the server; there's a ceiling on how much you can scale vertically. Downtime may be required during the upgrade process.



SCALING IN CLOUD COMPUTING (2)

Horizontal Scaling

(Add more same-size nodes)

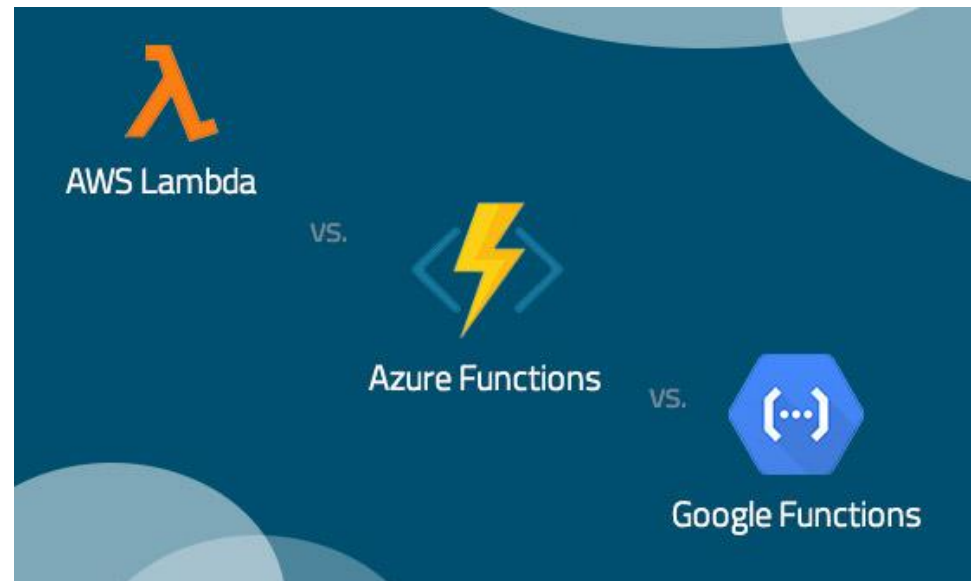


- **Horizontal Scaling (Scaling Out/In):** Involves adding or removing multiple servers or instances to handle increased load. For instance, deploying additional web servers behind a load balancer to distribute traffic.
 - **Advantages:** handle a much larger load by adding more instances. High Availability as the failure of one instance doesn't impact the overall application.
 - **Disadvantages:** Requires load balancing and can change to the application architecture to support distributed systems. More instances to monitor and manage can complicate operations.



SERVERLESS ARCHITECTURE

- A serverless architecture is a way to build and run applications and services without having to manage infrastructure. The application still runs on servers, but all the server management is done by cloud providers. Developers no longer have to provision, scale, and maintain servers to run their applications, databases, and storage systems.



CLOUD SERVICE PROVIDERS

Amazon Maintains Dominant Lead in the Cloud Market

Worldwide market share of leading cloud infrastructure service providers in Q3 2024*



- A cloud service provider (CSP) is a company that offers on-demand computing resources, which could be data storage, computing power, applications, infrastructure, and platform.



CLOUD SERVICE PROVIDERS

Feature / Service Area	AWS	Azure	GCP	IBM Cloud	Oracle Cloud	Alibaba Cloud
Compute	EC2, Lambda	Virtual Machines, Azure Functions	Compute Engine, Cloud Functions	Virtual Servers, Code Engine	Bare Metal, OCI Functions	ECS, Function Compute
Storage	S3, EBS, Glacier	Blob Storage, Disk Storage	Cloud Storage, Persistent Disks	Cloud Object Storage	Object & Block Storage	OSS, NAS
Database	RDS, DynamoDB, Aurora	SQL Database, Cosmos DB	Cloud SQL, Firestore, Spanner	Db2, Cloudant	Autonomous DB, MySQL	ApsaraDB (RDS, NoSQL)
Networking	VPC, CloudFront, Route 53	Virtual Network, Load Balancer	VPC, Cloud Load Balancing	VPC, Direct Link	FastConnect, VCN	VPC, Express Connect
AI / ML	SageMaker, Comprehend	Azure ML, Cognitive Services	Vertex AI, AutoML	Watson AI	AI Services	Machine Learning Platform for AI
Big Data & Analytics	Redshift, EMR, Kinesis	Synapse Analytics, HDInsight	BigQuery, Dataflow	IBM Analytics Engine	Oracle Analytics Cloud	MaxCompute, DataWorks
Security	IAM, KMS, Shield	Azure AD, Key Vault, Defender	IAM, Cloud Armor	Identity & Access, Key Protect	IAM, Vault	RAM, Cloud Firewall, KMS
Hybrid Cloud	Outposts	Azure Arc	Anthos	Red Hat OpenShift, Satellite	Oracle Cloud@Customer	Alibaba Cloud Stack
Global Reach	32+ regions	60+ regions	40+ regions	10+ regions	40+ regions	30+ regions
Best For	Most services, maturity	Enterprises & Microsoft users	Data & ML workloads	AI & regulated industries	Enterprise DBs & legacy apps	Asia-Pacific market focus



WHAT IS AWS?

- **Amazon Web Services (AWS)** is the world's leading cloud computing platform developed by Amazon.
- https://www.youtube.com/watch?v=a9_D53WsUs&ab_channel=AmazonWebServices



AWS GLOBAL INFRASTRUCTURE (1)

- AWS Global Infrastructure refers to the physical and virtual components that deliver Amazon Web Services (AWS) across the globe. It ensures high availability, low latency, security, and scalability for cloud applications.
- A Region is a geographically distinct area in the world where AWS has multiple data centers grouped together.
 - Each region contains multiple Availability Zones.
 - Each region is isolated from others for fault tolerance and data residency.
 - As of now, AWS has 30+ regions worldwide.
 - Example: us-east-1 (Northern Virginia), ap-southeast-1 (Singapore)



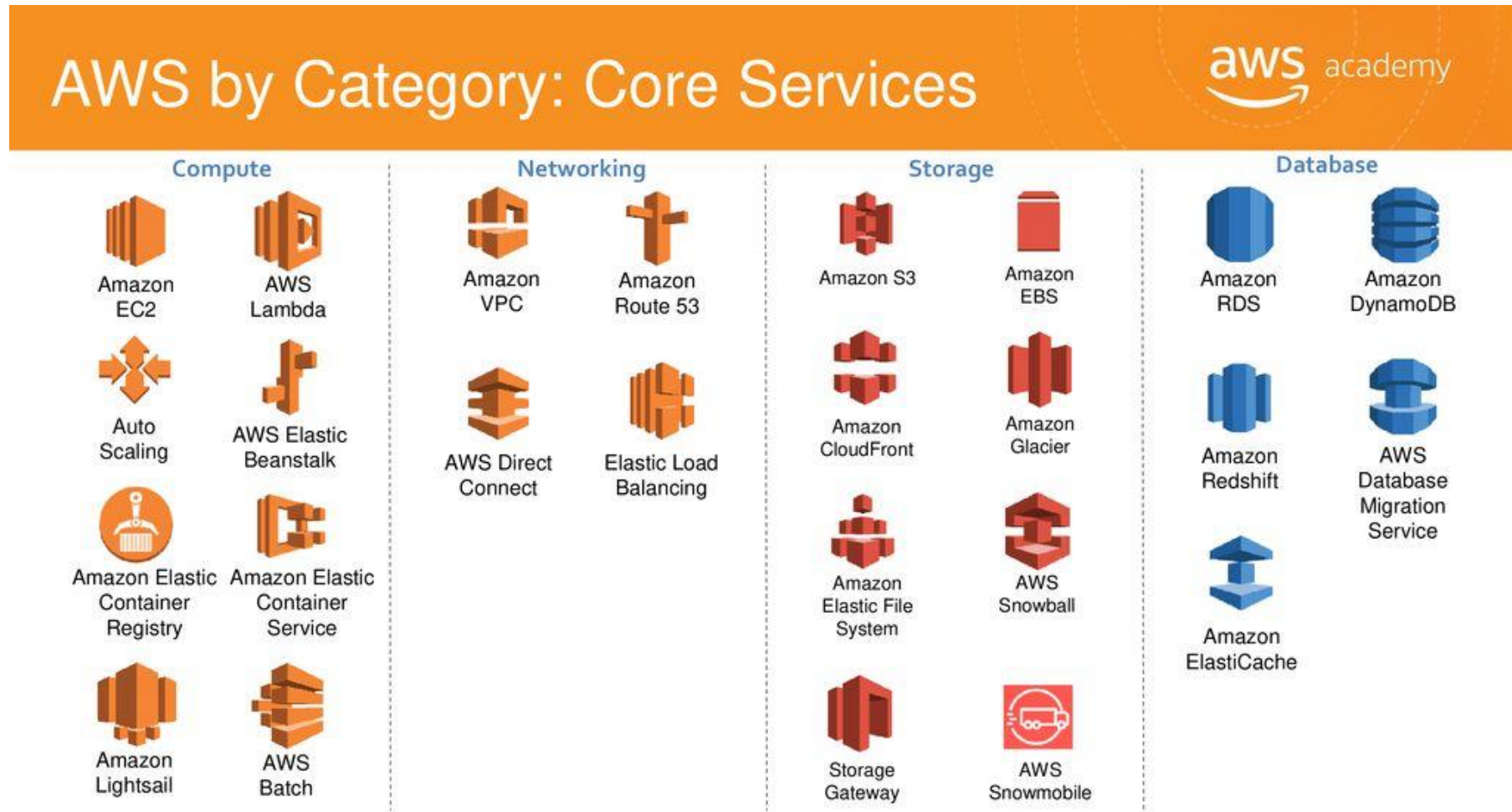
AWS GLOBAL INFRASTRUCTURE (2)



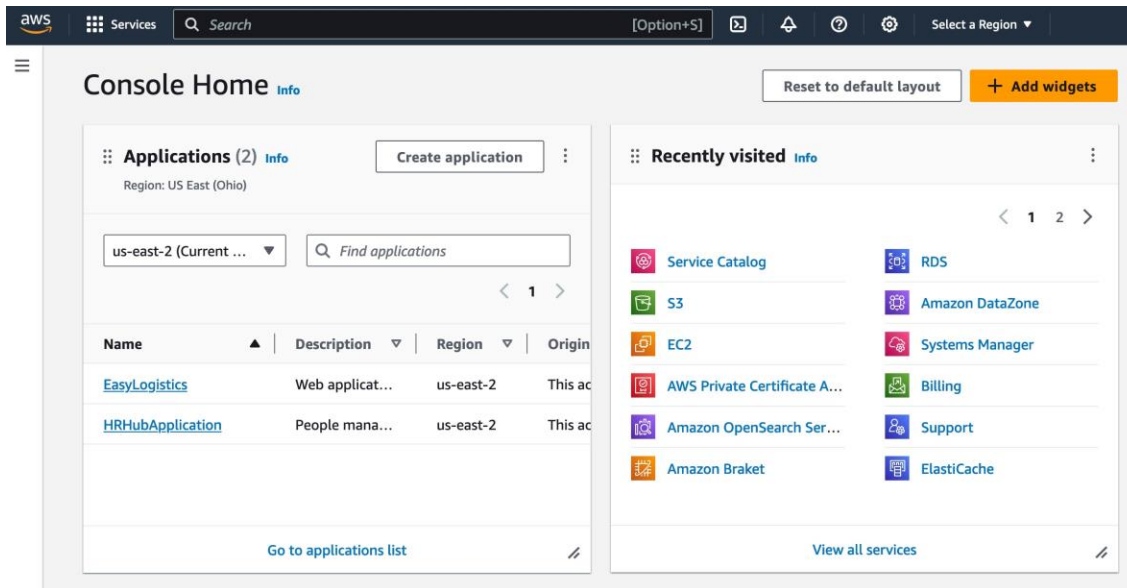
- Availability Zone is one or more discrete data centers within a Region.
 - Each AZ has independent power, cooling, and networking.
 - AZs are physically separated, often miles apart, to avoid single points of failure (but close enough for low latency).
 - Each Region contains 2 or more AZs.
 - An AZ is one or more physically separated data centers within a region.
 - Designed for high availability—you can run redundant apps across AZs.
 - Example: us-east-1a, us-east-1b, etc.



AWS CORE SERVICES



ACCESS TO AWS SERVICES



- AWS management console
- AWS command line interface (CLI)
- Infrastructure as Code (IaC):
Terraform, AWS Cloudformation,
AWS CDK
- AWS Software Development Kits (SDK):
 - AWS provides SDKs for various programming languages (e.g., Python, Java, JavaScript, Ruby).



AMAZON EC2

- <https://www.youtube.com/watch?v=TsRBftzZsQo>
- **Amazon Elastic Compute Cloud (Amazon EC2)** provides on-demand, scalable computing capacity in the Amazon Web Services (AWS) Cloud.
- An EC2 instance is a virtual server in the AWS Cloud.
- When you launch an EC2 instance, the instance type that you specify determines the hardware available to your instance.
- Each instance type offers a different balance of compute, memory, network, and storage resources.



Q & A

